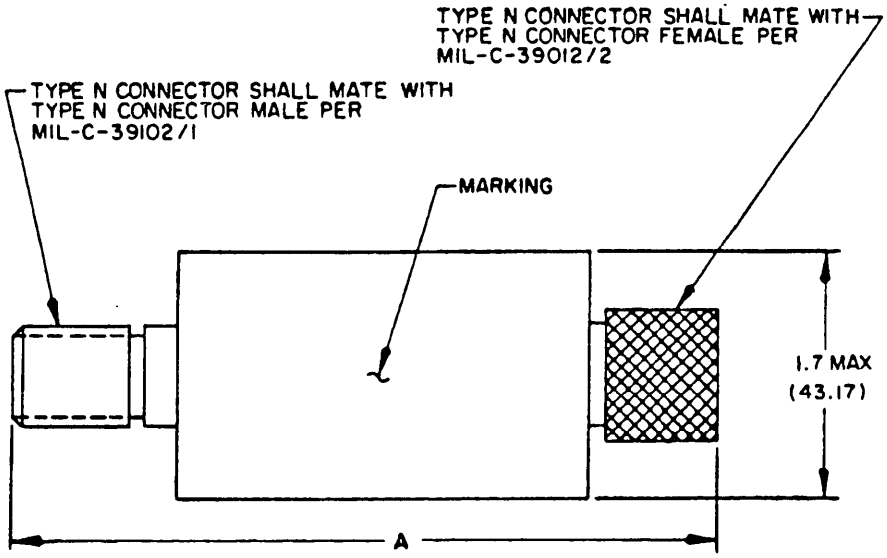
MILITARY SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (SERIES N),

FREQUENCY RANGE: DC TO 1.5 GHz, CLASS III, MEDIUM POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the attenuators described herein shall consist of this specification and the latest issue MIL-A-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

ENGINEERING DATA:

Class: III.

Operating frequency range: DC to 1.5 gigahertz (GHz).

Test frequencies for VSWR and attenuation measurements: DC, 0.5, 1.0, and 1.5 GHz.

Basic test frequency: 1.0 GHz.

Operating temperature range: -55°C to +85°C.

Maximum weight: 1 pound, 8 ounces.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I.

Power: See table I.

VSWR: See table I.

Power sensitivity: To be determined with 10 watts of average power.

Part number: M3933/3 (and dash number from table I).

Supersession data: See table II.

TABLE I. Electrical characteristics.

Dash number	Attenuation dB				Maximum power input		1/ VSWR maximum			Dimension A MAX
	Nominal	Deviation			AV (W)	Peak 2/ (kW)	DC	DC to 1.0 GHz	1.0 to 1.5 GHz	
		DC	DC to 1 GHz	1.0 to 1.5 GHz						
08 N S	1	±0.02	±0.2	±0.5	35	10	1.05:1	1.15:1	1.30:1	6.42 (163.07)
09 N S	2	±0.02	±0.2	±0.5	18	10	1.05:1	1.15:1	1.30:1	6.42 (163.07)
10 N S	3	±0.02	±0.2	±0.5	15	10	1.05:1	1.15:1	1.30:1	6.42 (163.07)
11 N S	4	±0.02	±0.2	±0.5	12	10	1.05:1	1.15:1	1.30:1	6.42 (163.07)
12 N S	5	±0.02	±0.2	±0.5	10	10	1.05:1	1.15:1	1.30:1	6.42 (163.07)
13 N S	6	±0.02	±0.2	±0.5	10	10	1.05:1	1.15:1	1.30:1	6.42 (163.07)
14 N S	10	±0.02	±0.2	±0.5	10	10	1.05:1	1.15:1	1.30:1	6.42 (163.07)
15 N S	20	±0.05	±0.5	±1.0	10	10	1.05:1	1.15:1	1.30:1	6.85 (173.99)

1/ VSWR value is for both ends.
2/ Peak power for a duty cycle of 5×10^{-4} maximum duration 5 microseconds.

TABLE II. Supersession data. 1/

Part number M3933/3-	Supersedes M3933/
10	3-5 and 11-1
13	11-2
14	11-3
15	11-4

1/ Part numbers M3933/3-1, -2, -3, -4, -6 and -7 (CN-121B/U, CN-158B/U, CN-159B/U, CN-160B/U, CN-331A/U, and CN-332A/U) were deleted from this specification previous revision; there were no requirements for these items.

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NOTE: Attenuators covered by this specification sheet are substitutable for the following manufacturer's or user's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

TABLE II. Substitution data.

Military part number M3933/3-	Manufacturer's code	Manufacturer's part number	NSN
08	93459	10-1	---
09	93459	10-2	---
10	93459	10-3	5985-00-439-6606
11	93459	10-4	---
12	93459	10-5	---
13	93459	10-6	5985-00-721-0193
14	93459	10-10	5985-00-721-0255
15	93459	10-20	---

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 99
DLA - ES

User activities:
Army - SM
Navy - MC, SH, AS

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0993-3)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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